DOE SBIR/STTR FY26 PHASE I · GENESIS MISSION

Topic 1, Scaling the Biotechnology Revolution

Copy each answer into the matching open-text field in AMP. Word counts are shown so you can check them against the Pitch Development Guide before pasting. Images and video are not enabled, so all four answers are plain text.

TOTAL WORDS	QUESTION 1	TOPIC
693 across four answers	**98** held under 100	**Locked** after the pitch is submitted

PROJECT TITLE TO ENTER IN AMP

Physics-Informed Machine Learning for Design of Continuous Immobilized-Enzyme Reactors Converting Post-Consumer PET to Fuel-Grade Ethanol

01 Summary, Topic, and Mission Alignment

98 WORDS

PlastiBioFuel is building a route from post-consumer PET to fuel-grade ethanol using a PET hydrolase immobilized in a covalent organic framework. The continuous spiral-flow reactor was designed in simulation: flow, channel geometry, and reusability modeled across more than a million runs. Wet-lab work at UNT confirms the enzyme and immobilization chemistry the design depends on, and identifies mass transfer as the limit. Transport is what reactor geometry decides. Phase I closes the gap between the model and hardware, turning it into a design tool with quantified error. That is AI applied to bioreactor design and biomanufacturing, Topic 1.

02 Technical Promise

199 WORDS

Significance. Enzymatic PET recycling is limited by reactor engineering, not enzyme discovery. Conditions are chosen one factor at a time, so little of the design space gets searched.

Innovation. FAST-PETase was reported by the Alper group at UT Austin in 2022, itself produced by machine learning. PlastiBioFuel did not invent the enzyme. Our contributions, covered by USPTO provisional 63/848,456, are the framework immobilization system, the spiral-flow geometry arrived at computationally, and the integrated process. Phase I moves machine learning up one level, from residue selection to reactor operation.

Feasibility. Three UNT milestones completed ahead of schedule, in a final report signed by Prof. Shengqian Ma. Free FAST-PETase breaks PET to terephthalic acid, confirmed by HPLC against standards. It loads into TAPB-BPDA-COF with no protein detectable in supernatant at 280 nm. Immobilized it stays active at a diminished rate: 122 µmol/h/mg turnover at 24 hours against a 246 to 348 literature range for free enzyme. UNT attributes the gap to mass transfer, substrate reaching the active site down the COF channel, and to insufficient pre-treatment. Transport is a geometry and flow problem, which is what the model covers. Phase I quantifies where it is trustworthy and ranks the experiments that close the gap.

> OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 26 WORDS
> Phase I ends with a measured number rather than a claim: quantified uncertainty on the model's reactor-scale predictions, and a ranked experiment sequence to close it.

03 Commercialization Potential

199 WORDS

Value proposition. The intended product is fuel-grade ethanol from a waste feedstock. Post-consumer PET is abundant, domestically sourced, priced well below agricultural feedstock, and it does not compete with food production. Low carbon fuel standard credit value applies to the finished fuel and is the firmer part of the value case. Federal renewable identification number eligibility remains subject to final renewable-fuel pathway qualification, and we do not assume it.

Competitive advantage. Pyrolysis carries high energy cost; mechanical recycling degrades polymer every cycle. Enzymatic recyclers returning monomer to virgin-equivalent resin sell into the resin market, so our offtake channel does not overlap theirs. Software vendors modeling bioprocess do not hold the chemistry; we have wet-lab confirmation of the immobilization and enzyme behavior our design depends on. Wet-lab groups iterate hardware one build at a time; our geometry was searched in simulation first. Holding both shortens the bench-to-plant step, where enzymatic routes stall.

Go to market. We are in offtake discussions with ARG Petro, a regional fuel distributor, and have their interest. Texas puts PET waste aggregation, refining infrastructure, and fuel blending demand in one region. The Phase I output is a sizing and operating envelope for the first modular unit, so the model feeds a capital decision.

> OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 23 WORDS
> A first modular unit sited near a Texas PET aggregation point is the specific commercial target that Phase I is meant to de-risk.

04 Team Qualifications

197 WORDS

Company. Michael David Ward, Founder and CEO, is the principal investigator and is primarily employed by PlastiBioFuel. He built the project's simulation and modeling capability and runs it himself, so the computational work Phase I extends stays inside the company rather than moving to a subcontractor. PlastiBioFuel is a service-disabled veteran-owned small business verified through SBA VetCert and holds USPTO provisional 63/848,456. Its laboratory program produced the preliminary data cited here. All work will be performed in the United States.

Partnerships. The University of North Texas is our research partner under executed Sponsored Research Agreement SRA1016, with a $250,000 extension under negotiation. Dr. Shengqian Ma, Department of Chemistry, leads covalent organic framework and immobilization chemistry; his group has published on enzyme immobilization in covalent organic frameworks. Dr. Joshua Phipps, postdoctoral researcher, runs day-to-day wet lab work. The partnership gives Phase I reactor access, HPLC analytics, and materials characterization with no capital spending.

Follow-on funding. PlastiBioFuel has a separate NSF SBIR Phase I proposal, Research.gov 328095, submitting against the November 4, 2026 window. That proposal covers wet chemistry and reactor hardware and does not overlap the computational scope proposed here. We disclose it so DOE can confirm there is no duplicate funding.

> OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 19 WORDS
> Marshall Nadel is a seed investor in PlastiBioFuel, and Patrick Tarlton of the Texas Concrete Association supports business development.

WORD COUNT SUMMARY

As written · With add-on

01. Summary, Topic, and Mission Alignment	98 words as written · 98 with add-on
02. Technical Promise	199 words as written · 225 with add-on
03. Commercialization Potential	199 words as written · 222 with add-on
04. Team Qualifications	197 words as written · 216 with add-on
Total	693 words as written · 761 with add-on

Question 1 is held under 100 words. If the Pitch Development Guide sets higher limits, append the add-on sentences. If it sets lower limits, the answers cut cleanly at paragraph boundaries. Drop the Feasibility paragraph from Question 2 last, since it carries the preliminary data that makes the pitch credible.

OPTIONAL UPLOAD: BIBLIOGRAPHY AND REFERENCES CITED

> Prepared and included in this package. Worth uploading: it substantiates the statement that PlastiBioFuel did not invent the enzyme, and it shows the UNT partner's published record in enzyme immobilization in covalent organic frameworks.

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